

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Second Semester of B. Tech. Examination (CL/ME/EE)

May 2014

CL102.01 Environmental Sciences

Date: 16.05.2014, Friday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q-1 (a) Define the term Environment and Write guiding principle of environmental studies. [03]
- Q-1 (b) Explain the impact of technology and development on environment. [04]
- Q-2 (a) Explain Man Environment relationship in detail. [04]
- Q-2 (b) Briefly describe the sources and effects of noise pollution. [05]
- Q-2 (c) Explain structure of atmosphere with neat sketch in detail. [05]

OR

- Q-2 (a) Give general classification of common air pollutants and also Explain their sources briefly. [04]
- Q-2 (b) Explain the term "Environmental Degradation" and discuss the role of advanced technology in the degradation of environment. [05]
- Q-2 (c) Explain Component parts of Environment and explain Biosphere in detail. [05]
- Q-3 (a) Explain effects of air pollution on plant and material. [04]
- Q-3 (b) Describe sources and effects of water pollutants. [05]
- Q-3 (c) Describe the main causes of depletion of ozone layer and briefly comment over control measures taken to prevent further ozone depletion at international level. [05]

OR

- Q-3 (a) Give classification of water pollutants and describe organic water pollutants. [04]
- Q-3 (b) Enlist all current global environmental problems and discuss green house effect in details. [05]
- Q-3 (c) Define Eutrophication, its causes and effects. [05]

SECTION - II

- Q-4 (a) Define Acid Rain. Also mention environmental effects of acid rain. [05]
Q-4 (b) Differentiate between conventional & Non conventional Energy Source. [02]
Q-5 (a) What do you understand by biogeochemical cycle? Explain the oxygen cycle with neat sketch. [07]
Q-5 (b) Explain the significance of studying food chain. [04]
Q-5 (c) Explain concept of energy flow in ecosystem. [03]

OR

- Q-5 (a) Give classification of ecology & explain in detail structure of an ecosystem. [07]
Q-5 (b) Write in Detail: (i) Pond Ecosystem (ii) Forest Ecosystem [04]
Q-5 (c) Define following terms: Autecology, Heterotrophs, Denitrification [03]
Q-6 (a) List out the advantages & disadvantages of Solar & Wind Energy Sources. [05]
Q-6 (b) What is meant by Ecological Pyramid? Describe the pyramid of biomass for terrestrial ecosystem. [05]
Q-6 (c) (i) In ecosystem although the inorganic nutrients are recycled, the flow of energy is not. True/ False? [04]
(ii) Biogeochemical cycles are not affected by human activities. True/false?
(iii) Biogeochemical cycles move elements between biotic & abiotic environment. True/False?
(iv) Write chemical equation for photosynthesis process.

OR

- Q-6 (a) Enlist the problems due to over exploitation of energy resources & explain them in detail. [05]
Q-6 (b) Write objectives of Ecology. [05]
Q-6 (c) Explain concept of food web considering an example of food web of forest ecosystem. [04]